The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP5008
Subject Title	Microelectronics Packaging and Reliability
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with knowledge of materials and techniques for integrated circuit (IC) packaging, an understanding of different failure mechanisms, the study of testing techniques, and reliability requirements for the assembly and packaging of ICs.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the standard manufacturing processing for IC packaging and the knowledge of the use of various packaging materials; (b) explain different up-to-date packaging techniques for the current IC production; (c) analyze and study various types of failure mechanisms of IC due to the influence of packaging; (d) perform qualified testing on the IC packaging; (e) explain the limitations of different packaging techniques on the electrical performance of the IC; and (f) understand the general methodologies on the design and simulation of electronics packages/modules in the industry.
Subject Synopsis/ Indicative Syllabus	**Packaging materials and manufacturing processing**: Review materials/encapsulants use for IC packaging and the related manufacturing processing. **State-of-the-art packaging techniques**: Ball grid array, flip-chip technology, surface mount technology, etc. **Failure mechanisms, sites, and modes**: Classification of failure mechanisms. Analysis of failures. Failure accelerators. Models for failure mechanisms. Ranking of potential failures. Comparison of plastic and ceramic package failure modes. **Qualification and accelerated testing**: Qualification Process. Tailoring to the Application. Accelerated testing. Qualification tests. Continuous qualification. Industry practices. **Effects of packaging on the electrical performance**: Heat dissipation issues, mechanical stress issues, passivation issues, and hermeticity issues. **Modelling and simulation of high-power/high-frequency electronics packages/modules:** Electrical modelling and simulation. Thermal modeling and simulation. Structural modeling and simulation. Design for Reliability.
Teaching/Learning Methodology	**Lecture**: The fundamentals of reliability and packaging of various integrated circuits will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. **Tutorial**: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	f
1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓	✓	✓
2. Examination	60%	✓	✓	✓	✓	✓	✓
Total	100 %						

Continuous assessments consist ofassignments and mid-term test. These will assess the students' understanding of basic concepts and principles in packaging technologies.

The examination will be conducted to make a comprehensive assessment of students' intended learning outcomes as stated above.

Student Study Effort Expected

Class contact:	
▪ Lecture	33 Hrs.
▪ Tutorial	6 Hrs
Other student study efforts:	
▪ Reading and self-study	81 Hrs.
The total student study effort	120 Hrs.

Reading List and References

Milton Ohring and Lucian Kasprzak, Reliability and Failure of Electronic Materials and Devices, 2nd Edition, Elsevier B.V., 2011 (https://www.sciencedirect.com/book/9780120885749/reliability-and-failure-of-electronic-materials-and-devices).

Andrea Chen and Randy Hsiao-Yu Lo, Semiconductor Packaging Materials Interaction and Reliability, 1st Edition, CRC Press, 2012 (https://www.taylorfrancis.com/books/oa-mono/10.1201/b11260/semiconductor-packaging-andrea-chen-randy-hsiao-yu-lo).

Haleh Ardebili, Jiawei Zhang, and Michael Pecht, Encapsulation Technologies for Electronic Applications, 2nd Edition, Elsevier B.V., 2018 (https://www.sciencedirect.com/book/9780128119785/encapsulation-technologies-for-electronic-applications).

Yong LIU, Power Electronic Packaging: Design, Assembly Process, Reliability and Modelling, Springer Science & Business Media, 2012. (https://link.springer.com/book/10.1007/978-1-4614-1053-9)